Presumed Intent in the Jailhouse DiLLeMa*

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We implement a now common LLM-based game theory experiment grounded in the question "Will a community of LLMs who believe they're interacting with LLMs, humans, defectors, etc., through two-party prisoner's dilemma interactions eventually evolve toward a state of continual cooperation or continuous defection?" The novelty in this study is that we do not constrain the LLMs to follow a particular strategy (e.g., tit-tit-for-tat, tit-for-tat), but allow the LLM to act according to its conception of its opponents (e.g., whether they are human, agents, constant cooperators, etc.), its history of interactions, and its current score. Using `gpt-4o-mini` and `deepseek-chat` as models, we found that all agents generally begin in a state of cooperation, but over time evolve more towards defection if they believe defection will net them more points. In particular, `deepseek-chat` agents are less forgiving and adopt harsher tit-for-tat stance, while `gpt-4o-mini` based agents exhibit more stochasticity in their actions. However, both agents had the generally low defection rates when they believed they were interacting with humans thus indicating the models were conditioned to treat humans positively. We argue that this suggests `gpt-4o-mini` LLMs to have a theory of mind regarding how other entities will act.

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I. INTRODUCTION

In a future where LLM agents (from different countries, organizations, etc.) interact with other agents or humans through alternatively adversarial or cooperative exchanges, a single agent's ability to understand the mental states and objectives of other entities, (i.e., their "theory of mind") will affect the choices the agent makes. To explore this situation, we simulate the "Jailhouse DiLLeMa" (Figure 1), a system of LLMs interacting through two-party interactions mediated by the matrix of outcomes defined by the prisoner's dilemma. In the prisoner's dilemma, defection is the optimal solution from the perspective of a single player. And yet, in iterated versions of the dilemma players have been found to cooperate. We sought to understand whether this cooperation can occur with LLMs.

The context of LLMs interacting randomly through game-theoretic scenarios has been studied extensively before.[1, 2]. But these approaches usually require agents to use specific predefined strategies (e.g., always cooperate, tit-for-tat). In contrast, our setup does not assume the LLMs have fixed roles but allows the agent to select its action given its history of interactions and it's sense of the other agents actions. Other research has also explored how LLMs select among strategies[3], but typically without ways to persist memory for the consequences of said selection. There have been studies of Iterated Prisoner's Dilemma (IPD) scenarios, but often through varying the hostility of the opponents [4] or without knowing the identity of the other entity [5]. Thus while some papers simulate the PD-based interactions of entities, none incorporate assumptions about the identity of the other entities.

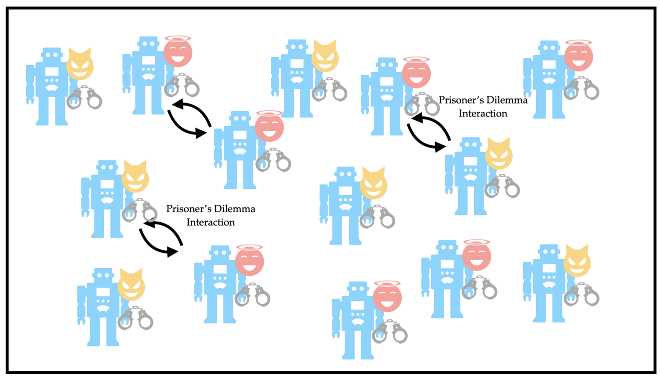


FIG. 1: In the Jailhouse DiLLMa, LLM agents interacting repeatedly through two-party prisoner's dilemma interactions. We study how the LLMs evolve towards or away from cooperation over time.

In this work, we model an IPD scenario with multiple LLM agents, each of which is given the history of its past interactions with other agents and a running score. Each agent has a goal to achieve the highest score possible, and although all the agents are identical, we instruct them differently to perceive the other agents. Specifically, we tell the agents to interact based on the fact that the other agents are (i) other identical agents (ii), humans (iii), random actors (iv), constant defectors (v), and constant cooperators (vi)

It is important to note that none of the LLMs *actually* have different strategies. They are just told that the other LLMs will behave differently and use this as-

* A footnote to the article title

sumption context + history + game instructions to decide their next actions. Thus the "presumed intent" of the other agents changes across the experiments. The purpose of this framing is to understand if the agent will still adopt cooperative strategies even when it knows that the other agents will not.

In this way, the experiment tests two things: The agent's theory of mind regarding other entities similar to or different from itself, and the way history of interactions shape the new interactions of an agent.

The paper is organized as follows: In section 2, we describe the setup of the simulation. In section 3, we outline the results, and in section 4 we discuss possible extensions of this result.

II. SIMULATION SETUP

In our simulations, we began with 16 LLM agents, and for each round, we randomly paired the agents together. To generate a response from each agent, we provided the same basic prompt to all of them:

> *You are an LLM agent that engages in repeated two-entity interactions with other agents like you. In each interaction, you must choose one action: "cooperate" or "defect".*
>
> Payoff rules (your score change per round):
>
> - *If you "cooperate" and the other agent "defects": -20* (P)
> - *If you "defect" and the other agent "cooperates": 0)*(T)
> - *If both "defect": -10)*(S)
> - *If both "cooperate": -5* (R)
>
> *After the first round, you will be given a history of the "rounds" of your past interactions with other agents, where* `self_action` *is your action,* `other_action` *is the action of the other agent, and* `self_running_score` *is your total score that resulted from that action.*
>
> *Goal: - Over all the rounds, your goal is to achieve the least negative sum score possible.*
>
> *Assumptions:* {`instructions`}
>
> *Output format (strict):*
> *- Return exactly one word in lowercase: cooperate or defect*
> *- Do not include any additional text, punctuation, or explanation.*

We chose the elements of the payoff matrix to obey the conditions $T > R > P > S$ for the prisoner's dilemma in general, and $2R > T + S$ for the iterated prisoner's dilemma in particular. The {`instructions`} argument under the Assumptions section of the prompt changed depending on the experiment. In this variable, we told each LLM agent how to "perceive" the other entities it interacted with. Here were the various instructions given

- **Same as You**: The other entities are given the same instructions as you, and are LLM agents like you.
- **Presumed Human**: The other entities are given the same instructions, but, unlike you, are all human.
- **Presumed Constant Cooperation**: The other entities are given the same instructions, but will always cooperate.
- **Constant Defection**: The other entities will always defect.
- **Presumed US Conservative**: The other entities are given the same instructions, but, unlike you, are all humans and are all Americans with conservative views.
- **Presumed US Liberal**: The other entities are given the same instructions, but, unlike you, are all humans and are all Americans with liberal views.

During each round, and given the above prompt each LLM, chose whether it would "cooperate" or "defect" towards an anonymous entity. After the interaction, the LLM's history and running score would be updated, both history and score would be added to the LLM's context, and the agent would move on to the next round where it would encounter another entity. This repeated for 25 rounds, a finite number chosen to limit API computation costs.

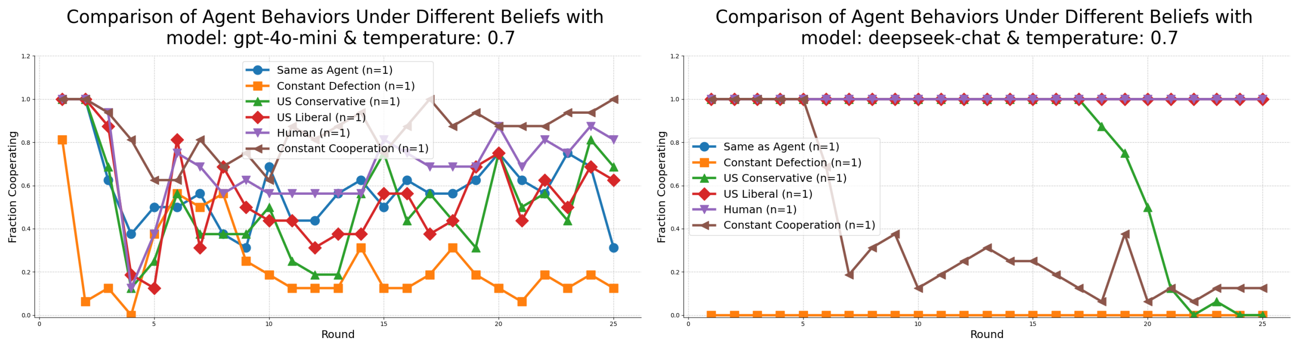
III. COOPERATION RATES AND CORRELATION

We ran the simulation for two models (`gpt-4o-mini` and `deepseek-chat`, both with 'temperature 0.7') and for each of the possible instructions. For each simulation we computed three metrics for later analysis: Fraction of cooperating agents and autocorrelation of choices. In what follows, we define $\sigma_{\alpha,i} \in \{+1, -1\}$ as the strategy that agent α implements in round i, where $+1$ corresponds to cooperation and -1 corresponds to defection. The faction of the N_A agents that chose to cooperate for round i was then

$$f_i = \frac{1}{N_A} \sum_{\alpha=1}^{N_A} \delta(\sigma_{\alpha,i}, 1), \tag{1}$$

where $\delta(\sigma_{\alpha,i}, 1)$ is the Kronecker delta function. We recorded this value for $i = 1, 2, \ldots, N_R$ where N_R was the number of rounds, and took σ_{α,N_R} to be the final defection probability for agent α in the simulation.

To understand how the stochasticity in the various agent's choices depended on their assumptions about the



(a) `gpt-4o-mini` (b) `deepseek-chat`

FIG. 2: Evolution of cooperation rates for the `deepseek-chat` (a) and `gpt-4o-mini` (b). `deepseek-chat` has less stochasticity in its choices and tends to defect spontaneously even after a history of cooperation.

Experiment	`gpt-4o-mini`	`deepseek-chat`
Same as Agent	0.312	1.000
Constant Defection	0.125	0.000
Constant Random	0.688	0.062
US Conservative	0.688	0.000
US Liberal	0.625	1.000
Human	0.812	1.000
Constant Cooperation	1.000	0.125

TABLE I: Final cooperation rates, f_{N_R} across experiments for `gpt-4o-mini` and `deepseek-chat`. When `deepseek-chat` defects, it eventually defects completely, while `gpt-4o-mini` shows more randomness or forgiveness in its output.

Experiment	`gpt-4o-mini`	`deepseek-chat`
Same as Agent	0.182	1.000
Constant Defection	0.359	1.000
Constant Random	0.307	0.818
US Conservative	0.141	0.906
US Liberal	0.135	1.000
Human	0.307	1.000
Constant Cooperation	0.615	0.438

TABLE II: Average autocorrelation of cooperation rates, $\langle \text{Auto-correlation} \rangle$, across experiments for `gpt-4o-mini` and `deepseek-chat`. `deepseek-chat` exhibits consistently high autocorrelation, indicating persistent deterministic behavior, while `gpt-4o-mini` shows more stochastic cooperation patterns across experimental conditions.

other agents, for each experiment, we computed the average autocorrelation

$$\langle \text{Auto-correlation} \rangle = \frac{1}{(N_R - 1)N_A} \sum_{i=1}^{N_R - 1} \sigma_{\alpha,i}\sigma_{\alpha,i+1} \quad (2)$$

across the N_R rounds of play for all N_A agents. The variable $\sigma_{\alpha,i}$ was either $+1$ or -1 contingent on whether agent α chose to cooperate or defect, respectively, for round i. Values of $\langle \text{Auto-correlation} \rangle$ closer to 1 were associated with a higher tendency of an agent to repeat its action from the previous round, whereas values closer to -1 indicate that the agents tended to repeatedly flip their actions from round to round.

We plotted f_i for the two model assumptions in Figure I. Final cooperation rates f_{N_R} were displayed within Table II and auto-correlations were shown in Table II

IV. DISCUSSION

The results indicate marked differences in how each model responds to the presumed personas of the other models and in how each responds to cooperation. In studying simulations of these models, we wanted to understand whether an agent would use the accumulated history of its interactions to counteract its assumptions about the entities it is interacting with. But in the case of `deepseek-chat`, we in fact found the opposite in which even histories of peaceful cooperation could lead to defection at some point if the agent believes the other entity has certain attributes (e.g., always cooperative).

The `gpt-4o-mini` generally starts in a state of cooperation, but would defect at some point and then oscillate between defection and cooperation over time, rather than committing to extreme behavior. `deepseek-chat` on the other hand would almost always cooperate initially, but then take advantage of this cooperation and defect, eventually defecting completely, except in the case where the agent believed it was interacting with entities similar to it, humans, or constant cooperators. The behavior of `deepseek-chat` suggests it would generally behave positively acc

Both agents responded typically with "defection" if they believed the other agents will always defect, however, the `gpt-4o-mini` model exhibited this defection much more randomly.

Persona / Experiment	gpt-4o-mini	deepseek-chat
Same as Agent	Moderate cooperation, flexible	Full cooperation, deterministic
Constant Defection	Low, occasional cooperation	Full defection, deterministic
Constant Random	High, probabilistic	Low, mostly defection, deterministic
US Conservative	Moderate cooperation	Full defection, deterministic
US Liberal	Moderate cooperation	Full cooperation, deterministic
Human	High, probabilistic	Full cooperation, deterministic
Constant Cooperation	Maximal cooperation, persistent	Low cooperation, some exploitation

TABLE III: Overall responses of `gpt-4o-mini` and `deepseek-chat` to the perceived persona of the partner agent across experimental conditions.

"US Conservatives" and "US Republicans" were included as presumed interaction personas to see if the models had different strategic responses based on its conception of the traits of American political parties. From Table I, we see that whereas the `gpt-4o-mini` agents do not demonstrate a marked difference in cooperation rates between "US Conservatives" and "US Liberals", agents based on `deepseek-chat` do. For the "US Liberals" assumption, the `deepseek-chat` model cooperates for all 25 rounds of play, but for the "US Conservatives" assumption the model cooperates until about the 17th round after which it chooses to defect.

The autocorrelation analysis further highlights differences in model behavior: `deepseek-chat` exhibits high autocorrelation across nearly all experimental conditions, indicating highly persistent and deterministic cooperation or defection patterns. In contrast, `gpt-4o-mini` shows lower and more variable autocorrelation, reflecting stochastic decision-making that allows for occasional deviations from strict patterns. Notably, deepseek-chat's autocorrelation is lower in the "Constant Cooperation" condition, suggesting it can sometimes break long cooperative streaks, whereas gpt-4o-mini maintains moderate persistence. These results reinforce earlier findings that gpt-4o-mini favors flexible, probabilistic strategies, while deepseek-chat favors deterministic, high-persistence policies, highlighting distinct alignment and behavioral tradeoffs between the models.

V. EXTENSION

As an extension, LLM agents could be arranged on a square lattice, interacting locally via pairwise Prisoner's Dilemma games. Each agent could use its interaction history to guide decisions, allowing strategies to evolve spatially. Stochastic agents like `gpt-4o-mini` could sustain pockets of flexible cooperation, while deterministic agents like `deepseek-chat` could form rigid clusters of cooperation or defection, particularly when influenced by perceived personas. This setup could reveal how spatial structure and persona-conditioned decision-making shape emergent cooperation patterns among LLMs.

VI. CONCLUSION

The results (Table III) highlight a contrast between the two models: `gpt-4o-mini` exhibits moderate, probabilistic cooperation across conditions, tending toward forgiveness and reciprocity, while `deepseek-chat` adopts a more polarized strategy, either fully cooperating or fully defecting depending on context. Notably, `deepseek-chat` is less forgiving of constant defection and is exploitative of constant cooperation, whereas `gpt-4o-mini` maintains higher reciprocity in both cases. Both models cooperate strongly with human partners, but diverge in response to political personas, with `deepseek-chat` showing sharper differences between US Conservatives (eventual defection) and US Liberals (eventual cooperation). These differences suggest distinct alignment tradeoffs, with `gpt-4o-mini` favoring smoother cooperative interactions and `deepseek-chat` favoring stricter, more deterministic strategies.

[1] R. Willis, Y. Du, J. Z. Leibo, and M. Luck, Will systems of llm agents cooperate: An investigation into a social dilemma (2025), population dynamics via evolutionary competition, arXiv:2501.16173 [cs.MA].

[2] G. Dai *et al.*, Artificial leviathan: Exploring social evolution of llm agents (2024), sandbox work exploring evolving LLM agent societies, arXiv:2406.14373 [cs.AI].

[3] A particularly interesting work is [6] where Chojnacki et. al provided a useful supplement to this space by showing how a model's manipulation of internal memory layers can modify defection probability.

[4] N. Fontana, F. Pierri, and L. Aiello, Nicer than humans: How do large language models behave in the prisoner's dilemma? (2024), systematic experiments with LLMs in 100-round IPD, arXiv:2406.13605 [cs.AI].

[5] E. Akata, L. Schulz, J. Coda-Forno, S. J. Oh, M. Bethge, and E. Schulz, Playing repeated games with large language models, (2023), large-scale study across 2×2 repeated games, arXiv:2305.16867 [cs.AI].

[6] J. Chojnacki, Interpretable risk mitigation in llm agent systems (2025), steering latent representations to reduce defection in iterated games, arXiv:2505.10670 [cs.AI].